

**RATIONAL POINTS ON A TWIST OF THE KLEIN QUARTIC AND
THE CLASSIFICATION OF 7-ADIC IMAGES OF GALOIS
REPRESENTATIONS FOR ELLIPTIC CURVES OVER $\mathbb{Q}$**

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ABSTRACT. We show that the modular curve $X_{E_3}(7)$, associated with $E_3 : y^2 = x^3 - x^2 - 16x + 29$, has exactly four rational points. We find a degree two elliptic quotient over a number field of degree 21, which yields 37-adic height equations vanishing at every rational point. We use these equations to check every residue disc at 37 and show that the four known rational points are the only ones. This proves Furio-Lombardo's Conjecture 1.6, which completes their classification of 7-adic Galois images for elliptic curves over $\mathbb{Q}$ without complex multiplication.